Main Figures

Parallel transcriptomics reveals insect developmental shifts and gall-associated host transcriptional differences in *Quadrastichus erythrinae* and *Erythrina variegata*

Serdar DİNÇ

Figures 1–6

Figure 1. Study system and experimental design

Figure 2. Insect GO and KEGG enrichment

Figure 3. Insect secretome and candidate families

Figure 4. Plant GO and KEGG enrichment

Figure 5. Plant-defense candidates

Figure 6. MEROPS inhibitors and peptidases

Figure 1

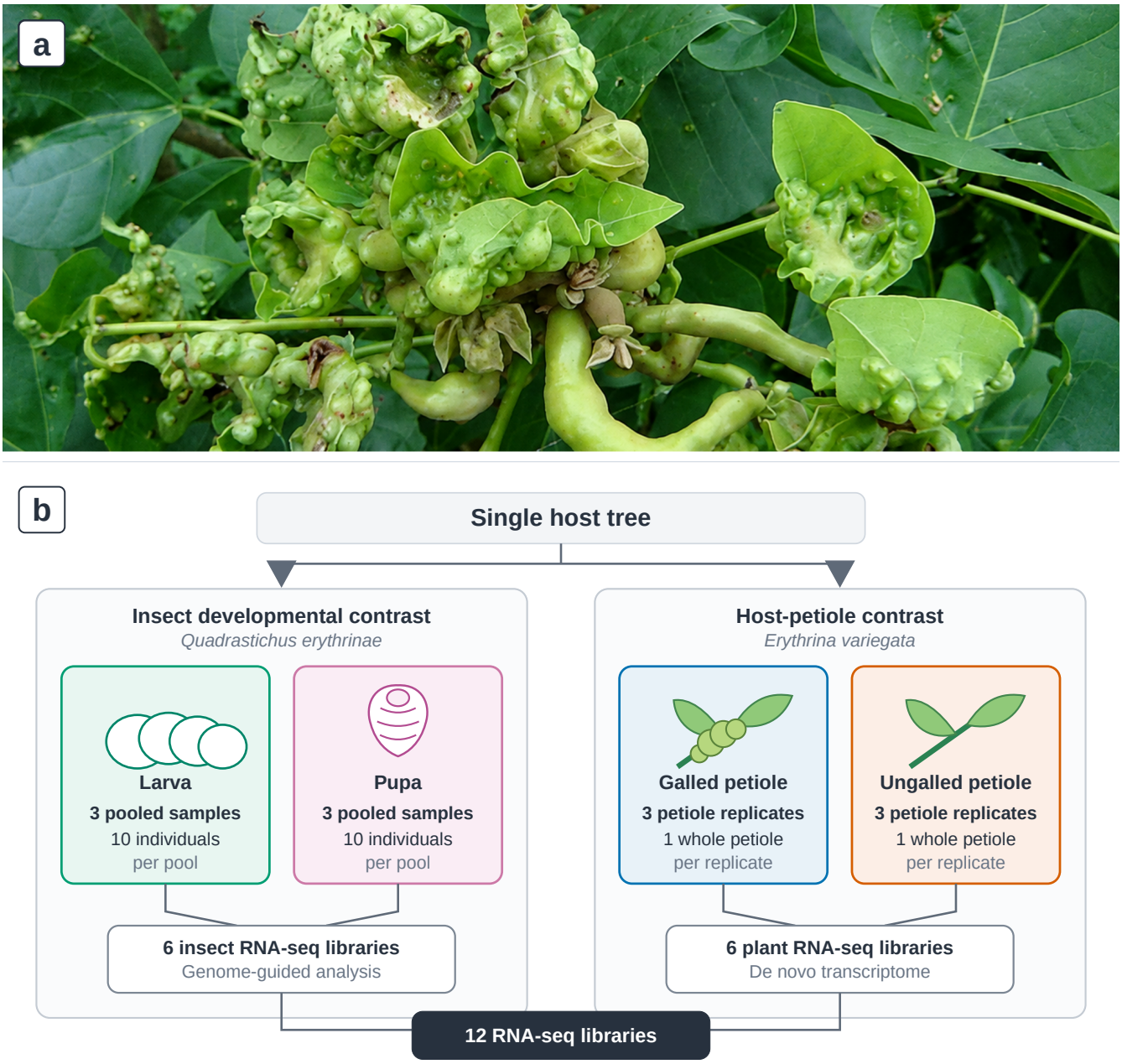


Fig. 1. Study system and experimental design. (a) *E. variegata* galls induced by *Q. erythrinae*. Image cropped from Lin et al. [9], Fig. 1B (CC BY 4.0). (b) Separate larva–pupa and galled–ungalled petiole comparisons from a single tree sampled once in Taipei, Taiwan, in November 2020. Each insect stage comprised three pools of 10 individuals from different galled petioles; each plant condition comprised three whole-petiole replicates (GP, galled; UP, ungalled).

Figure 2

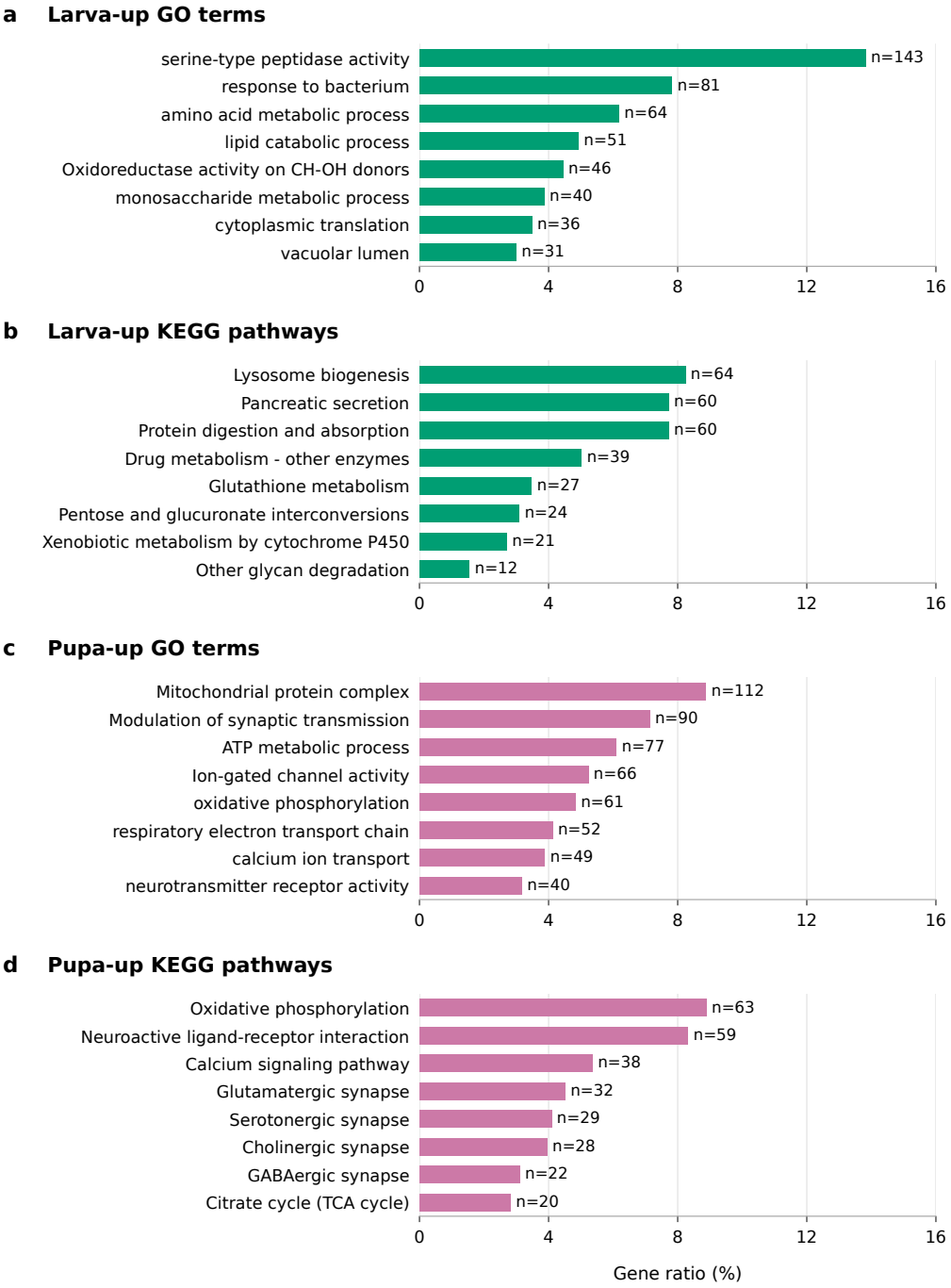


Fig. 2. Representative GO terms and KEGG pathways enriched among larva-up and pupa-up genes in *Q. erythrinae*. (a) Larva-up GO terms. (b) Larva-up KEGG pathways. (c) Pupa-up GO terms. (d) Pupa-up KEGG pathways. Larval and pupal categories are shown in green and pink, respectively. Bars show the percentage of GO- or KEGG-annotated genes in each directional gene set assigned to the indicated category; labels show gene counts. All displayed categories had BH-adjusted $P < 0.05$. Representative categories were selected to reduce redundancy, although categories may share contributing genes. Complete results are provided in Supplementary Tables S4 and S5.

Figure 3

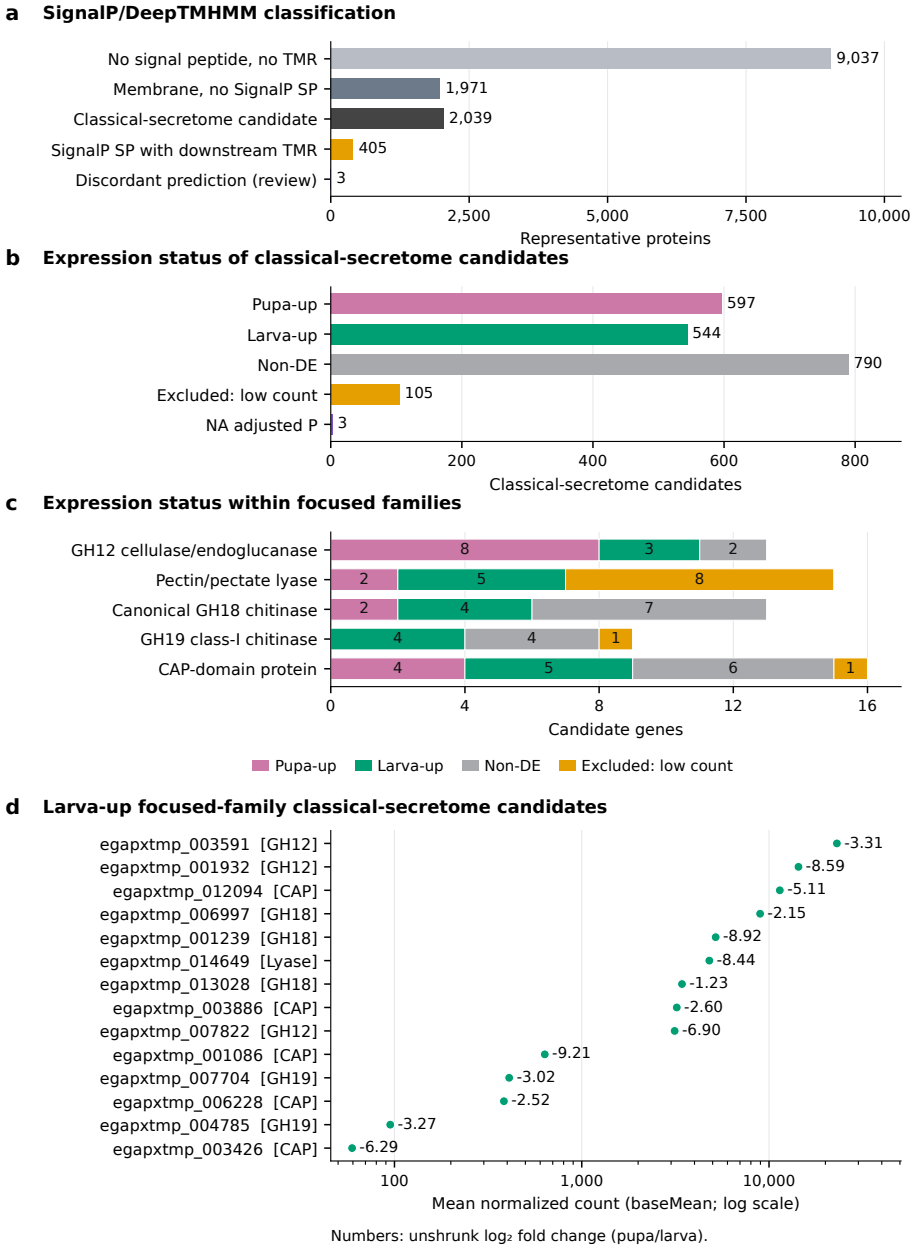


Fig. 3. Predicted classical-secretome and focused candidate-family profiles in *Q. erythrinae*. (a) Classification of 13,455 representative proteins using SignalP and DeepTMHMM. (b) Expression status of the 2,039 proteins meeting the conservative classical-secretome criterion: a predicted SignalP signal peptide and no predicted DeepTMHMM transmembrane region. (c) Expression status within five focused candidate-family inventories. (d) Mean normalized counts for the 14 focused-family genes that were both larva-up and predicted classical-secretome candidates. In (b) and (c), green and pink indicate larva-up and pupa-up genes, respectively. Points in (d) are uniformly green; bracketed abbreviations identify candidate families, and numerical labels show unshrunk $\log_2$ fold changes for pupa versus larva. DEG classification required BH-adjusted $P < 0.01$ and an absolute estimated, unshrunk $\log_2$ fold change ≥ 1. Non-DE genes did not meet both criteria; low-count exclusions and unavailable adjusted P values are shown separately. Classical-secretome classification predicts entry into the classical secretory pathway; extracellular localization and effector activity remain to be tested.

Figure 4

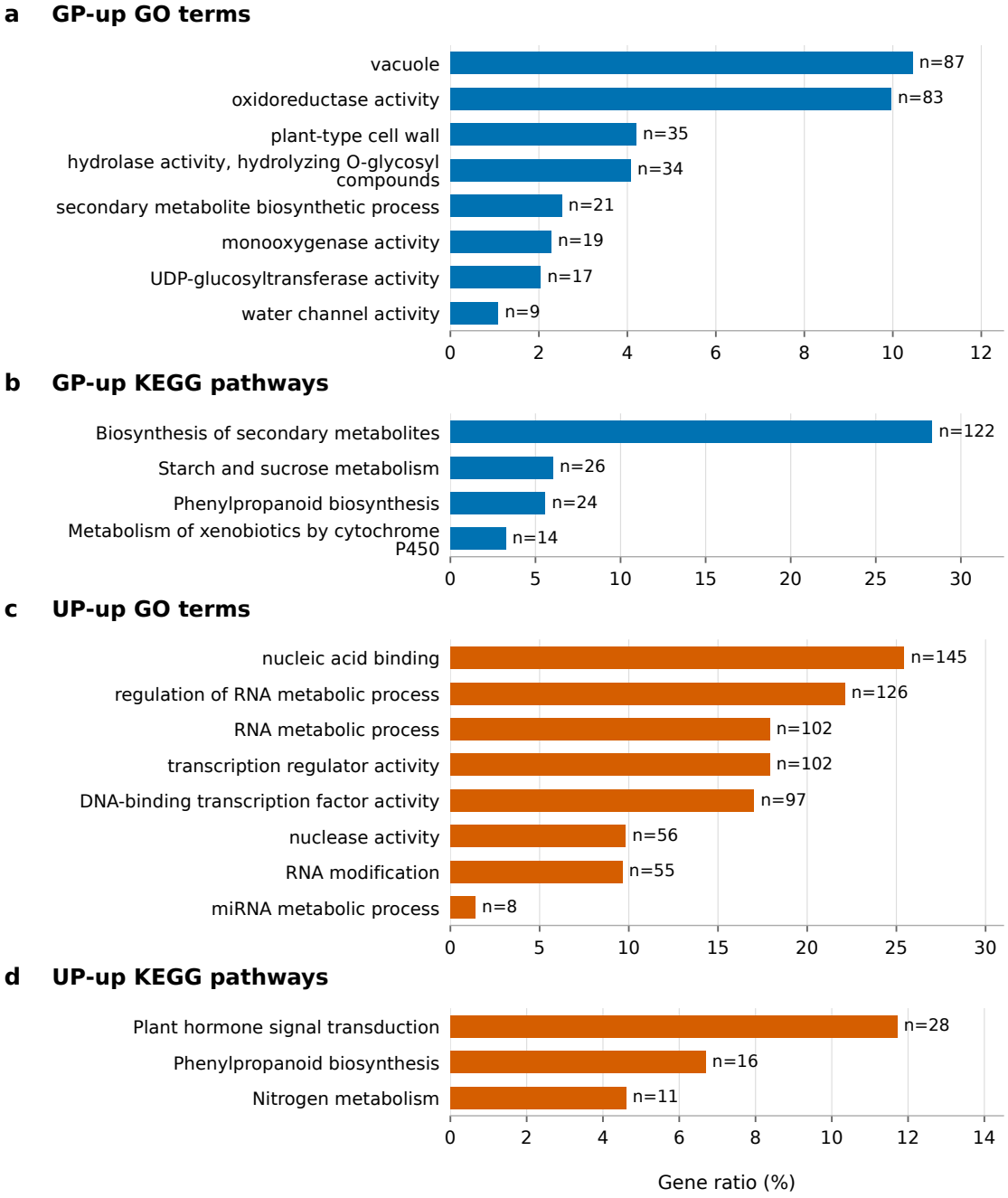


Fig. 4. Representative functional enrichment associated with differential expression in galled and ungalled *E. variegata* petioles. Currently resolved GO terms and significant KEGG pathways are shown for GP-up genes (a, b; blue) and UP-up genes (c, d; orange). Bars indicate the percentage of annotation-eligible Trinity gene clusters in each directional set assigned to a category, and labels show cluster counts. All displayed categories had BH-adjusted $P < 0.05$. Redundant GO terms were omitted from the figure but retained in Supplementary Table S11. The broad KEGG category metabolic pathways and the category chemical carcinogenesis-DNA adducts were omitted from panel b but retained in Supplementary Table S12. The latter shares all 14 GP-up contributing genes with metabolism of xenobiotics by cytochrome P450 and therefore does not provide an independent gene-level enrichment signal.

Figure 5

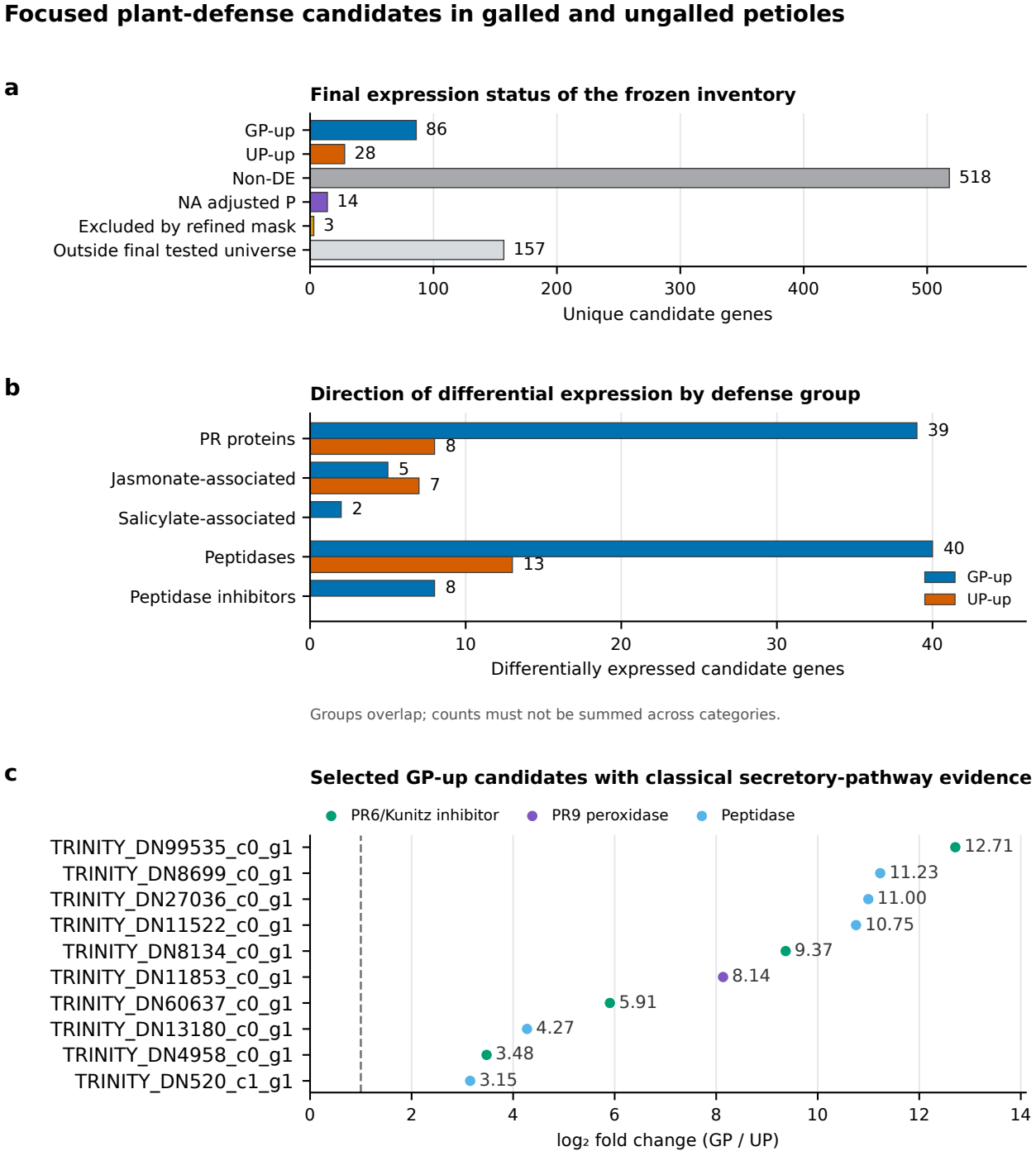
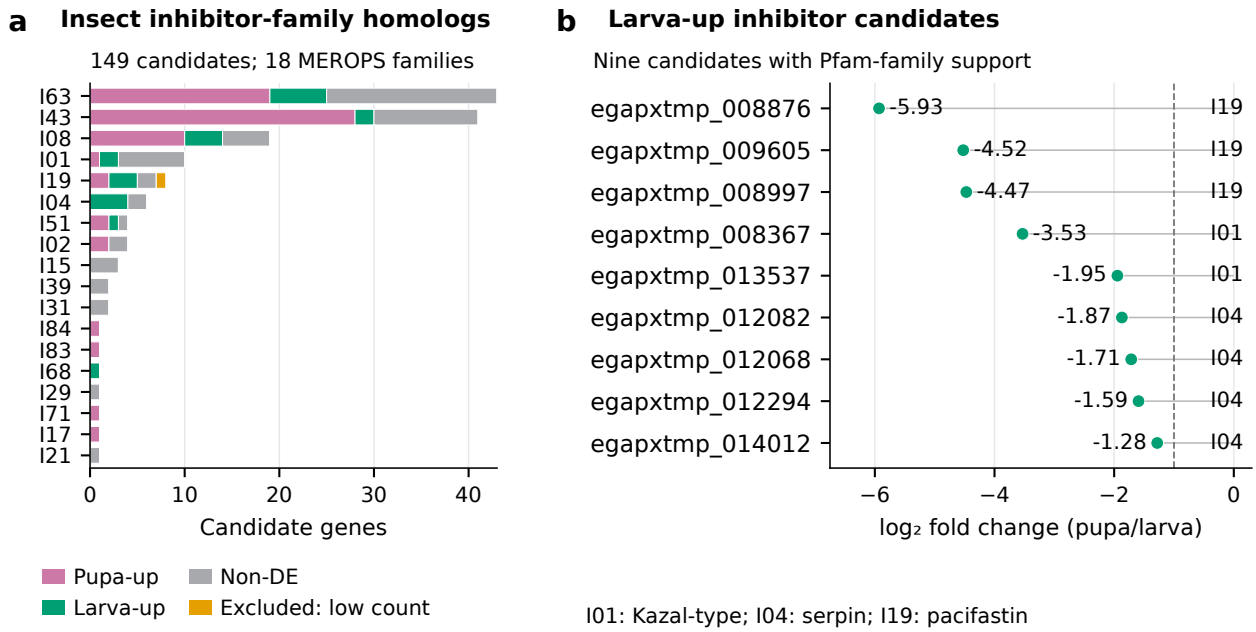
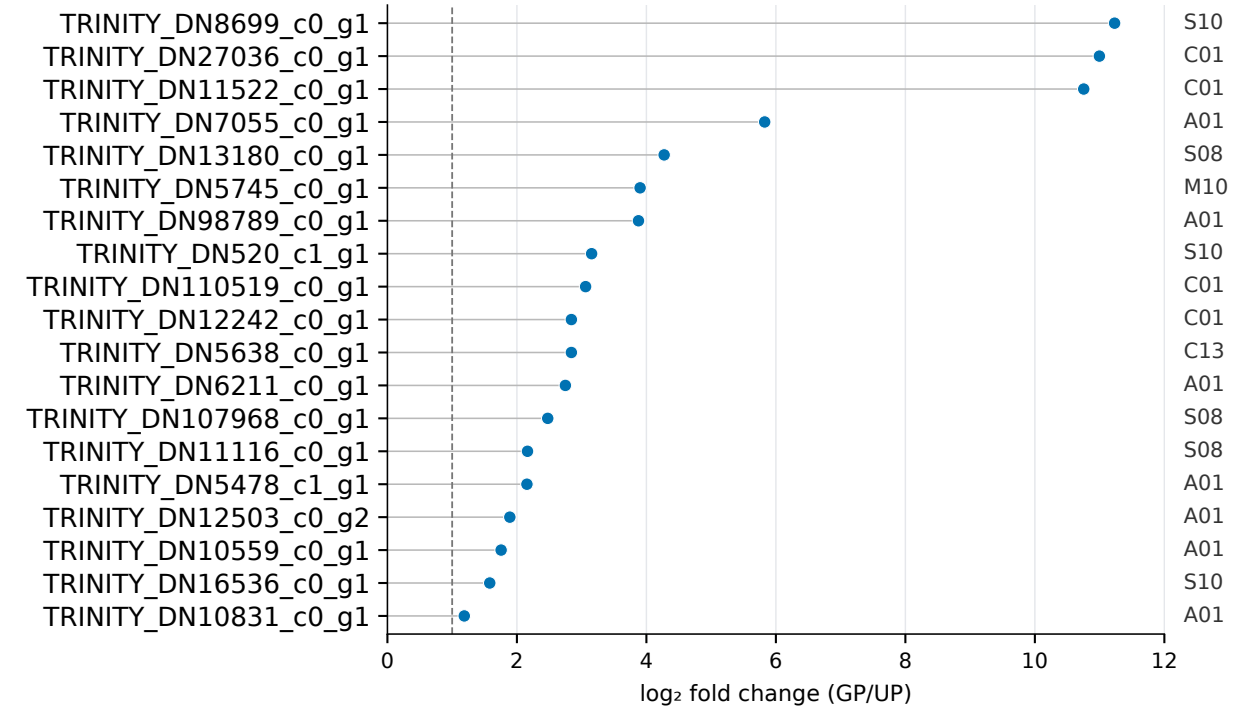

Focused plant-defense candidates in galled and ungalled petioles
a
Final expression status of the frozen inventory
GP-up
UP-up
Non-DE
NA adjusted P
Excluded by refined mask
Outside final tested universe
86
28
518
14
3
157
0
100
200
300
400
500
Unique candidate genes
b
Direction of differential expression by defense group
PR proteins
Jasmonate-associated
Salicylate-associated
Peptidases
Peptidase inhibitors
39
8
5
7
2
40
13
8
GP-up
UP-up
0
10
20
30
40
Differentially expressed candidate genes
Groups overlap; counts must not be summed across categories.
c
Selected GP-up candidates with classical secretory-pathway evidence
PR6/Kunitz inhibitor
PR9 peroxidase
Peptidase
TRINITY_DN99535_c0_g1
TRINITY_DN8699_c0_g1
TRINITY_DN27036_c0_g1
TRINITY_DN11522_c0_g1
TRINITY_DN8134_c0_g1
TRINITY_DN11853_c0_g1
TRINITY_DN60637_c0_g1
TRINITY_DN13180_c0_g1
TRINITY_DN4958_c0_g1
TRINITY_DN520_c1_g1
12.71
11.23
11.00
10.75
9.37
8.14
5.91
4.27
3.48
3.15
0
2
4
6
8
10
12
14
log₂ fold change (GP / UP)
All displayed candidates were SignalP-positive and lacked a predicted transmembrane region.

Figure 5 caption

Fig. 5. Expression patterns within the annotation-based plant-defense candidate inventory. Membership was frozen before the final refined-mask expression overlay. (a) Final expression status of all 806 candidate Trinity gene clusters. (b) Numbers of GP-up and UP-up candidates within five overlapping groups from this inventory. Group-specific counts are not additive because some candidates belong to multiple groups. (c) Ten selected GP-up predicted classical-secretome candidates from the inventory. Candidates were SignalP-positive and lacked a predicted transmembrane region. Points are colored by candidate class, and labels show unshrunk $\log_2$ fold changes for GP versus UP. The dashed line marks $\log_2$ fold change = 1. The topology predictions indicate entry into the classical secretory pathway; extracellular localization and biological activity remain untested.

Figure 6


a
Insect inhibitor-family homologs
149 candidates; 18 MEROPS families
I63
I43
I08
I01
I19
I04
I51
I02
I15
I39
I31
I84
I83
I68
I29
I71
I17
I21
0
10
20
30
40
Candidate genes
Pupa-up
Non-DE
Larva-up
Excluded: low count
b
Larva-up inhibitor candidates
Nine candidates with Pfam-family support
egapxtmp_008876
egapxtmp_009605
egapxtmp_008997
egapxtmp_008367
egapxtmp_013537
egapxtmp_012082
egapxtmp_012068
egapxtmp_012294
egapxtmp_014012
-5.93
-4.52
-4.47
-3.53
-1.95
-1.87
-1.71
-1.59
-1.28
I19
I19
I19
I01
I01
I04
I04
I04
I04
−6
−4
−2
0
log₂ fold change (pupa/larva)
I01: Kazal-type; I04: serpin; I19: pacifastin


c
GP-up plant peptidase candidates
Final refined-mask priority 1 + 2 classical-secretome candidates (n = 19)
TRINITY_DN8699_c0_g1
TRINITY_DN27036_c0_g1
TRINITY_DN11522_c0_g1
TRINITY_DN7055_c0_g1
TRINITY_DN13180_c0_g1
TRINITY_DN5745_c0_g1
TRINITY_DN98789_c0_g1
TRINITY_DN520_c1_g1
TRINITY_DN110519_c0_g1
TRINITY_DN12242_c0_g1
TRINITY_DN5638_c0_g1
TRINITY_DN6211_c0_g1
TRINITY_DN107968_c0_g1
TRINITY_DN11116_c0_g1
TRINITY_DN5478_c1_g1
TRINITY_DN12503_c0_g2
TRINITY_DN10559_c0_g1
TRINITY_DN16536_c0_g1
TRINITY_DN10831_c0_g1
S10
C01
C01
A01
S08
M10
A01
S10
C01
C01
C13
A01
S08
S08
A01
A01
A01
S10
A01
0
2
4
6
8
10
12
log₂ fold change (GP/UP)
MEROPS: A01 aspartic; C01 papain-like; C13 legumain/VPE; M10 metallo;
S08 subtilisin-like; S10 serine carboxypeptidase. All points: GP-up.

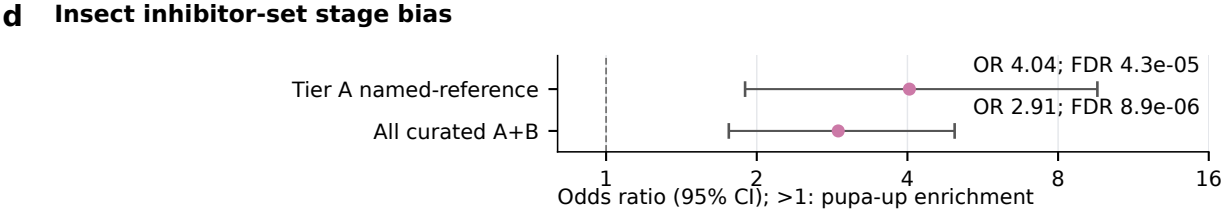

d
Insect inhibitor-set stage bias
Tier A named-reference
All curated A+B
OR 4.04; FDR 4.3e-05
OR 2.91; FDR 8.9e-06
1
2
4
8
16
Odds ratio (95% CI); >1: pupa-up enrichment

Figure 6 caption

Fig. 6. Stage-associated insect inhibitor-family homologs and prioritized GP-up plant peptidase candidates. (a) Expression-status distribution of 149 curated insect inhibitor-family homologs across 18 MEROPS families, selected from 2,039 predicted classical-secretome candidates after screening the complete representative proteome. (b) Nine of the 23 larva-up inhibitor-family homologs with the expected Pfam-family support. Negative $\log_2$ fold changes represent higher larval transcript abundance in the pupa-versus-larva contrast. (c) Nineteen of 20 priority plant peptidase candidates retained as GP-up after the final refined-mask analysis. The 20 candidates were curated from the original GP-up predicted classical-secretome search, and each represents a Trinity gene cluster. Positive $\log_2$ fold changes represent higher GP transcript abundance. Candidates in (b) and (c) are colored green and blue, respectively; adjacent codes identify MEROPS families. Stems connect estimates to zero and are not confidence intervals. (d) Pupa-up versus larva-up odds ratios for the curated inhibitor set (91 DEGs among 149 members) and Tier-A named-reference subset (47 DEGs among 64 members), compared with the remaining predicted classical-secretome DEGs (1,050 and 1,094, respectively). Points show odds ratios and horizontal lines show 95% confidence intervals. Dashed lines mark $\log_2$ fold changes of -1 in (b) and $+1$ in (c), and an odds ratio of 1 in (d). Non-DE genes did not meet both DEG criteria (BH-adjusted $P < 0.01$ and an absolute estimated, unshrunk $\log_2$ fold change ≥ 1); low-count exclusions and unavailable adjusted P values are separate categories. GP, galled petiole; UP, ungalled petiole. The independently selected insect and plant inventories do not identify interacting inhibitor–peptidase pairs.